

# NEXTGEN ANALYTICS

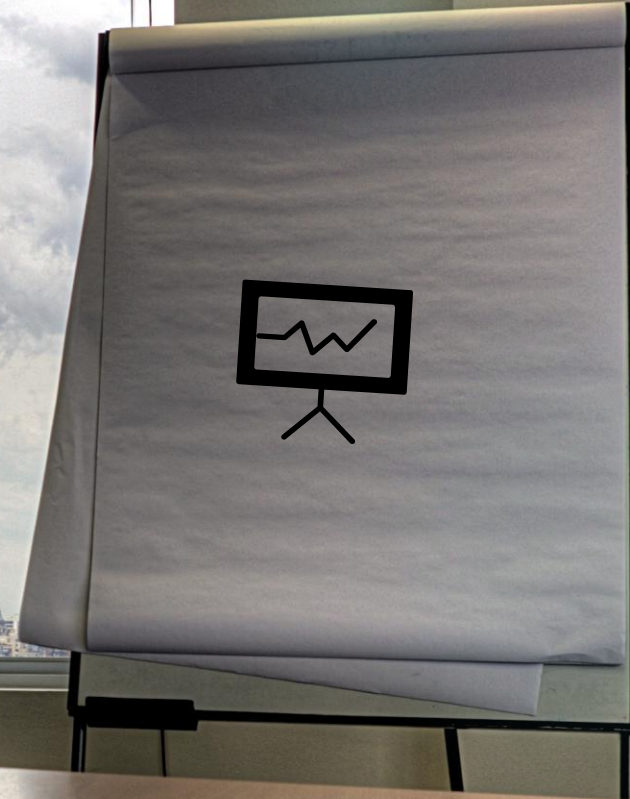
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## M8 Project Work: VISUALIZATIONS

TITLE :

# **Data-Driven Enforcement for Improved Housing Conditions and Code Compliance in NYC.**

*Welcome to our analysis of housing conditions and code compliance in New York City, where we leverage data to understand tenant experiences and identify critical areas for improvement.*

# Introduction

## Why Housing Code Compliance Matters in NYC

- **Brief Overview:**

- NYC faces widespread tenant complaints about housing conditions.
- Complaints often lead to violations, but **how effective is the response?**

- **Objective:**

- Use data to understand complaint trends.
- Identify borough-specific challenges and suggest data-driven solutions.

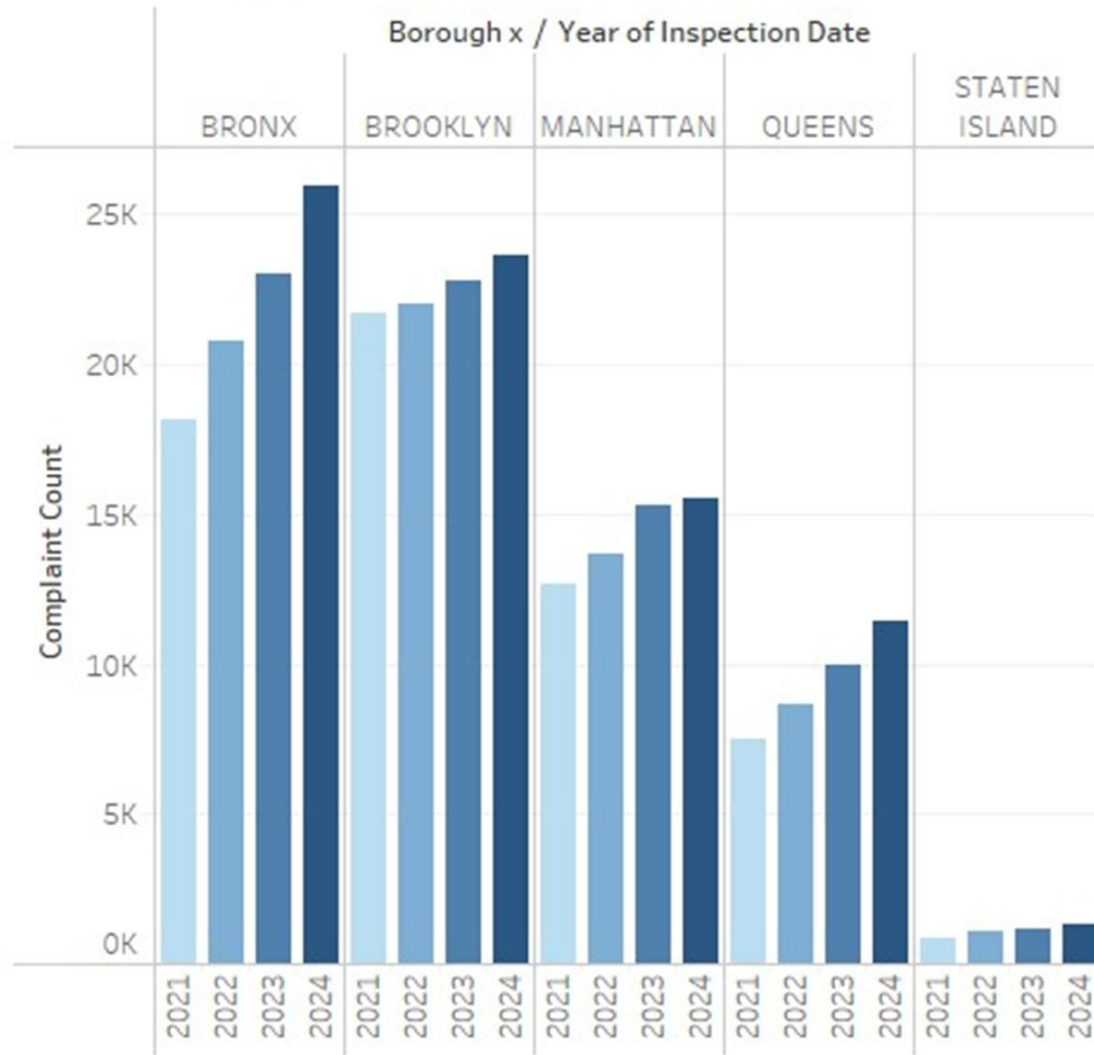
## Complaint Overview – Borough Trends

### Which Borough Faces the Most Housing Complaints?

#### Key Insights:

- Brooklyn has the highest complaint volume (449,391).
- The Bronx has the highest percentage of violations (46%)—indicating severity.
- Staten Island has the lowest complaint volume.

complaints across the years



## Breakdown by Complaint Type

### What Are the Most Common Housing Issues?

#### Key Insights:

- **Heat/Hot Water issues dominate** in Brooklyn (107,459 complaints) and the Bronx (81,103).
- **Unsanitary Condition issues** are also high in Brooklyn and the Bronx.
- **Queens has the lowest violation percentage** (21%), indicating **better resolution?**

| Borough       | Major Category |          |          |         |                   |          |        |                      |               | Grand Total |
|---------------|----------------|----------|----------|---------|-------------------|----------|--------|----------------------|---------------|-------------|
|               | APPLIAN..      | DOOR/W.. | ELECTRIC | GENERAL | HEAT/<br>HOT WA.. | PLUMBI.. | SAFETY | UNSANIT<br>ARY CON.. | WATER<br>LEAK |             |
| BROOKLYN      | 11,160         | 22,277   | 14,344   | 9,259   | 107,459           | 20,481   | 5,652  | 46,187               | 12,526        | 249,345     |
| BRONX         | 12,209         | 18,333   | 10,899   | 7,885   | 101,991           | 18,432   | 3,138  | 35,600               | 9,076         | 217,563     |
| MANHATTAN     | 6,918          | 14,248   | 6,653    | 8,751   | 81,103            | 14,112   | 2,568  | 29,263               | 7,547         | 171,163     |
| QUEENS        | 4,959          | 8,441    | 7,619    | 4,269   | 55,966            | 9,850    | 2,711  | 22,793               | 4,520         | 121,128     |
| STATEN ISLAND | 860            | 1,391    | 1,407    | 544     | 4,967             | 1,777    | 323    | 4,310                | 641           | 16,220      |
| Grand Total   | 36,106         | 64,690   | 40,922   | 30,708  | 351,486           | 64,652   | 14,392 | 138,153              | 34,310        | 775,419     |

## Do Complaints Lead to Violations?

### *How Many Complaints Turn into Violations?*

#### Key Insights:

- **Brooklyn has the highest complaint volume (449,391).**
- **The Bronx has the highest percentage of violations (46%)**—indicating severity.
- **Queens has the lowest violation percentage (21%)**

## Boroughs with the Most Complaints leading to violations

| Borough x     | complaint count | violation count | % of Complaints<br>Leading to Violations |
|---------------|-----------------|-----------------|------------------------------------------|
| BRONX         | 218,425         | 101,506         | 46%                                      |
| BROOKLYN      | 449,391         | 107,745         | 24%                                      |
| MANHATTAN     | 159,519         | 66,176          | 41%                                      |
| QUEENS        | 203,578         | 41,908          | 21%                                      |
| STATEN ISLAND | 20,785          | 5,061           | 24%                                      |
| Grand Total   | 1,051,698       | 322,396         | 31%                                      |

## Do Complaints Lead to Action?

### Are Complaints being resolved in time ?

#### Key Insights:

- **Bronx has the longest resolution times,** making it a hotspot for unresolved complaints.
- **Queens and Staten Island resolve complaints faster.**

## Average Time to Resolve Complaints

| Borough y     | Major Category | Complaint Count | Avg. average response time (days) |
|---------------|----------------|-----------------|-----------------------------------|
| BRONX         | WATER LEAK     | 9,076           | 12                                |
|               | PLUMBING       | 18,432          | 11                                |
|               | APPLIANCE      | 12,209          | 11                                |
|               | PAINT/PLASTER  | 20,989          | 11                                |
|               | HEAT/HOT WATER | 101,991         | 11                                |
|               | ELECTRIC       | 10,899          | 10                                |
| BROOKLYN      | WATER LEAK     | 12,526          | 11                                |
|               | PLUMBING       | 20,481          | 9                                 |
|               | APPLIANCE      | 11,160          | 9                                 |
|               | PAINT/PLASTER  | 19,032          | 9                                 |
|               | HEAT/HOT WATER | 107,459         | 8                                 |
|               | ELECTRIC       | 14,344          | 8                                 |
| MANHATTAN     | PLUMBING       | 14,112          | 11                                |
|               | WATER LEAK     | 7,547           | 10                                |
|               | PAINT/PLASTER  | 13,266          | 8                                 |
|               | APPLIANCE      | 6,918           | 8                                 |
|               | HEAT/HOT WATER | 81,103          | 8                                 |
|               | ELECTRIC       | 6,653           | 7                                 |
| QUEENS        | PLUMBING       | 9,850           | 7                                 |
|               | PAINT/PLASTER  | 7,601           | 7                                 |
|               | APPLIANCE      | 4,959           | 7                                 |
|               | ELECTRIC       | 7,619           | 6                                 |
|               | HEAT/HOT WATER | 55,966          | 6                                 |
|               | WATER LEAK     | 4,520           | 5                                 |
| STATEN ISLAND | PAINT/PLASTER  | 1,221           | 11                                |
|               | HEAT/HOT WATER | 4,967           | 8                                 |
|               | WATER LEAK     | 641             | 8                                 |
|               | PLUMBING       | 1,777           | 7                                 |
|               | ELECTRIC       | 1,407           | 6                                 |
|               | APPLIANCE      | 860             | 6                                 |

## How Quickly Are Violations Resolved?

### *Are Housing Violations Being Fixed Fast Enough?*

#### Key Insights:

- **High-priority violations (Class C) are resolved in ~25.5 days.**
- **Medium-priority violations take 35-42 days.**
- **Low-priority violations take nearly 3 months, with Brooklyn at **98.79 days** (longest).**

average response time (days) for each Violation Severity level Breakdown in each borough

| Class | severity label  | Borough |          |           |        |               |
|-------|-----------------|---------|----------|-----------|--------|---------------|
|       |                 | BRONX   | BROOKLYN | MANHATTAN | QUEENS | STATEN ISLAND |
| A     | Low Priority    | 96.33   | 98.79    | 96.30     | 96.40  | 96.07         |
| B     | Medium Priority | 42.83   | 35.00    | 40.61     | 36.39  | 36.11         |
| C     | High Priority   | 25.53   | 25.54    | 25.55     | 25.71  | 25.23         |

Long-Term Trends in Complaint Resolution

*Are Housing Complaints Being Addressed Faster Over Time?*

Key Insights:

- Closed complaints increased from 467,297 (2020) to 828,151 (2024).
- However, open complaints also peaked in 2024, suggesting a backlog.
- 2024 stands out as an outlier—possible policy shifts or operational changes?

Problem Status over the years

| Year of Problem Status D.. | Problem Status |        |
|----------------------------|----------------|--------|
|                            | CLOSE          | OPEN   |
| 2024                       | 828,151        | 14,908 |
| 2023                       | 746,411        | 358    |
| 2022                       | 715,624        | 232    |
| 2021                       | 606,990        | 165    |
| 2019                       | 595,575        | 75     |
| 2020                       | 467,297        | 73     |

## Borough-wise Analysis of Complaint Resolution Efficiency (days)

### Are Housing Complaints Being Addressed Faster Over Time?

#### Key Insights:

- Staten Island has the highest average compliance percentage (81%) indicating the most efficient resolution of complaints relative to expected timelines.
- The Bronx has the lowest average resolution time (35 days), suggesting the quickest handling of complaints, though their compliance percentage is the lowest (71%).
- Queens has the longest average expected resolution time (66 days) and the second-lowest compliance percentage (73%).
- All boroughs have an average resolution time lower than their average expected resolution time which shows that complaints are generally resolved faster than the initial estimates.
- Brooklyn has the second-longest average expected resolution time (65 days) and the second-lowest average resolution time (39 days).

## Borough-wise Analysis of Complaint Resolution Efficiency (days)

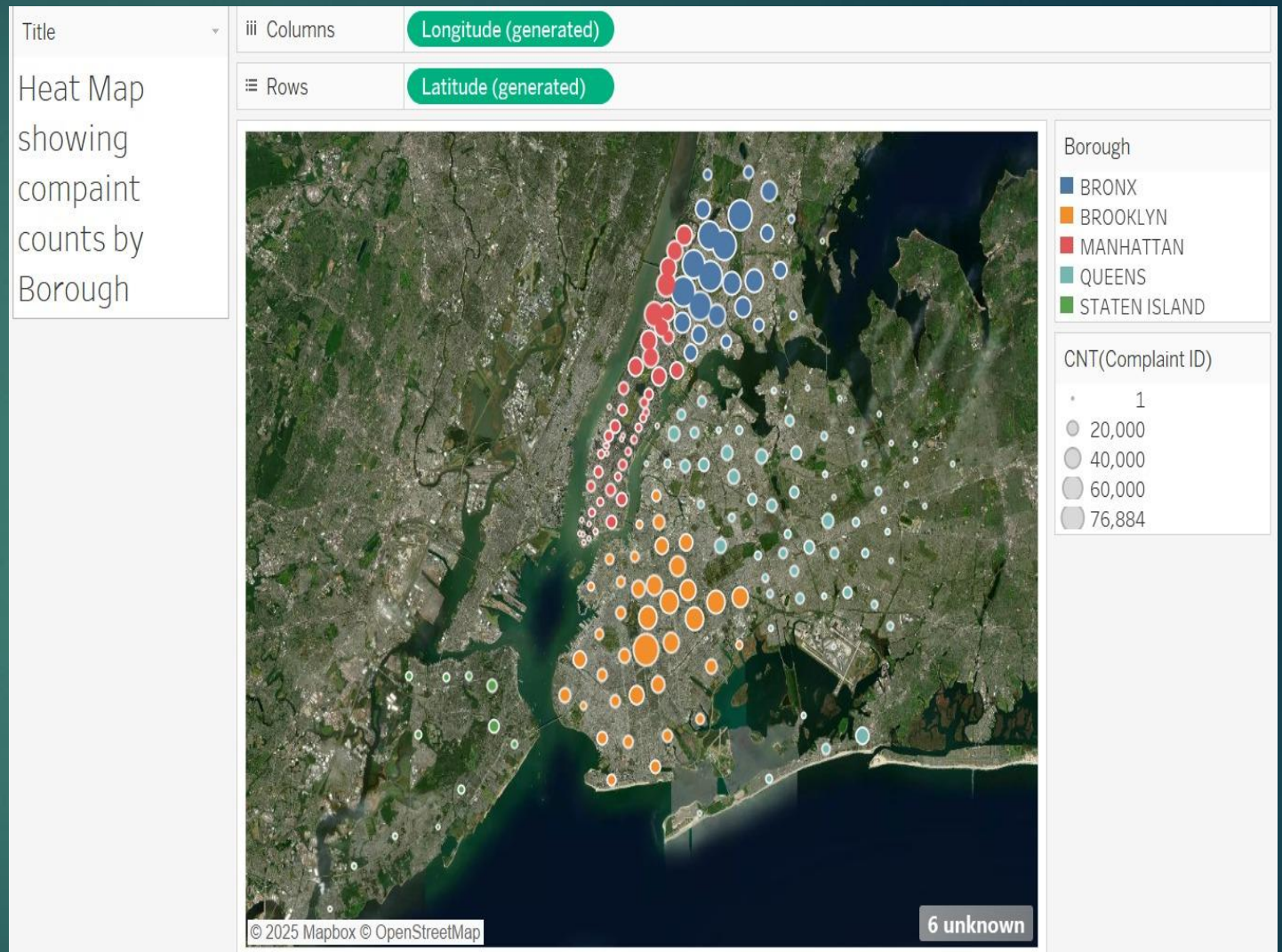
| Borough       | Avg. Expected resolution time | Avg. resolution time | Avg. compliance percentage |
|---------------|-------------------------------|----------------------|----------------------------|
| STATEN ISLAND | 58                            | 40                   | 81%                        |
| MANHATTAN     | 58                            | 39                   | 77%                        |
| BROOKLYN      | 65                            | 39                   | 74%                        |
| QUEENS        | 66                            | 37                   | 73%                        |
| BRONX         | 56                            | 35                   | 71%                        |

## Geographic Hotspots for Complaints

### *Where Are Housing Problems the Worst?*

#### Key Insights:

- Brooklyn & the Bronx have the most complaint hotspots.
- Geographic trends help prioritize interventions.



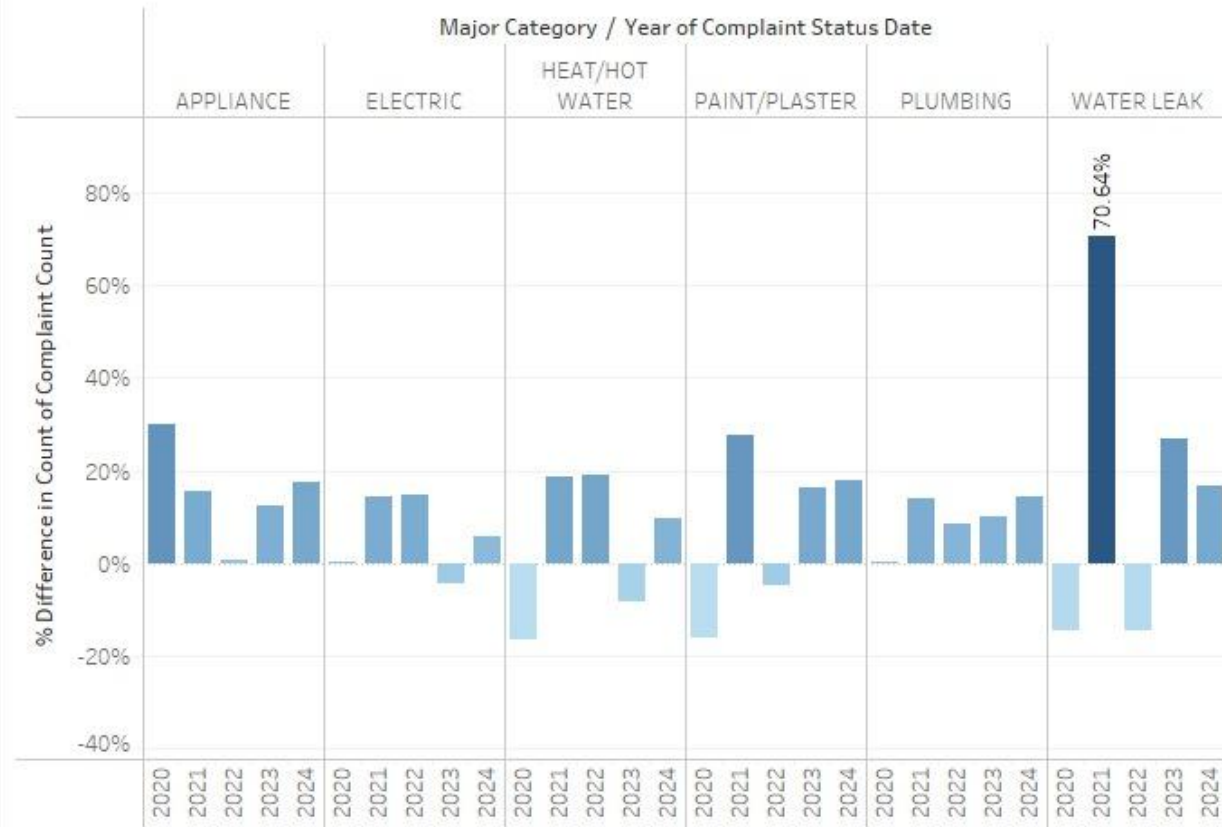
## Trends Over the Years

### *Are Some Housing Issues Getting Worse?*

#### Key Insights:

- Water leak complaints spiked in 2022.
- Heat/Hot Water complaints declined in 2021 but rose again after.
- Plumbing, electric, and appliance complaints show steady increases.

percentage change of complaint count each year

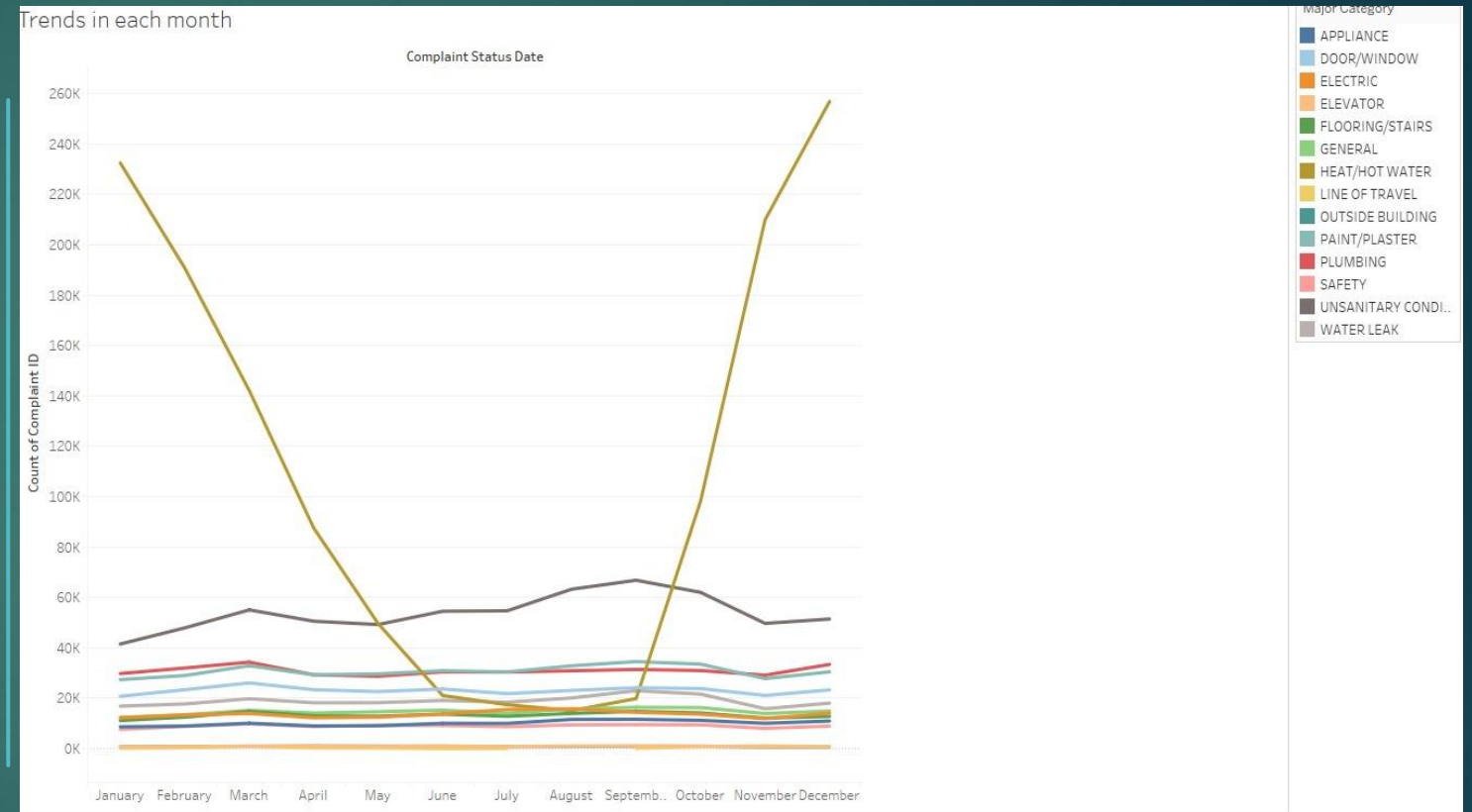


## Monthly Trends in Complaints

### Do Complaints Change with the Seasons?

#### Key Insights:

- **Heat/Hot Water complaints peak in December & January** (cold months).
- **Other complaint types remain steady.**

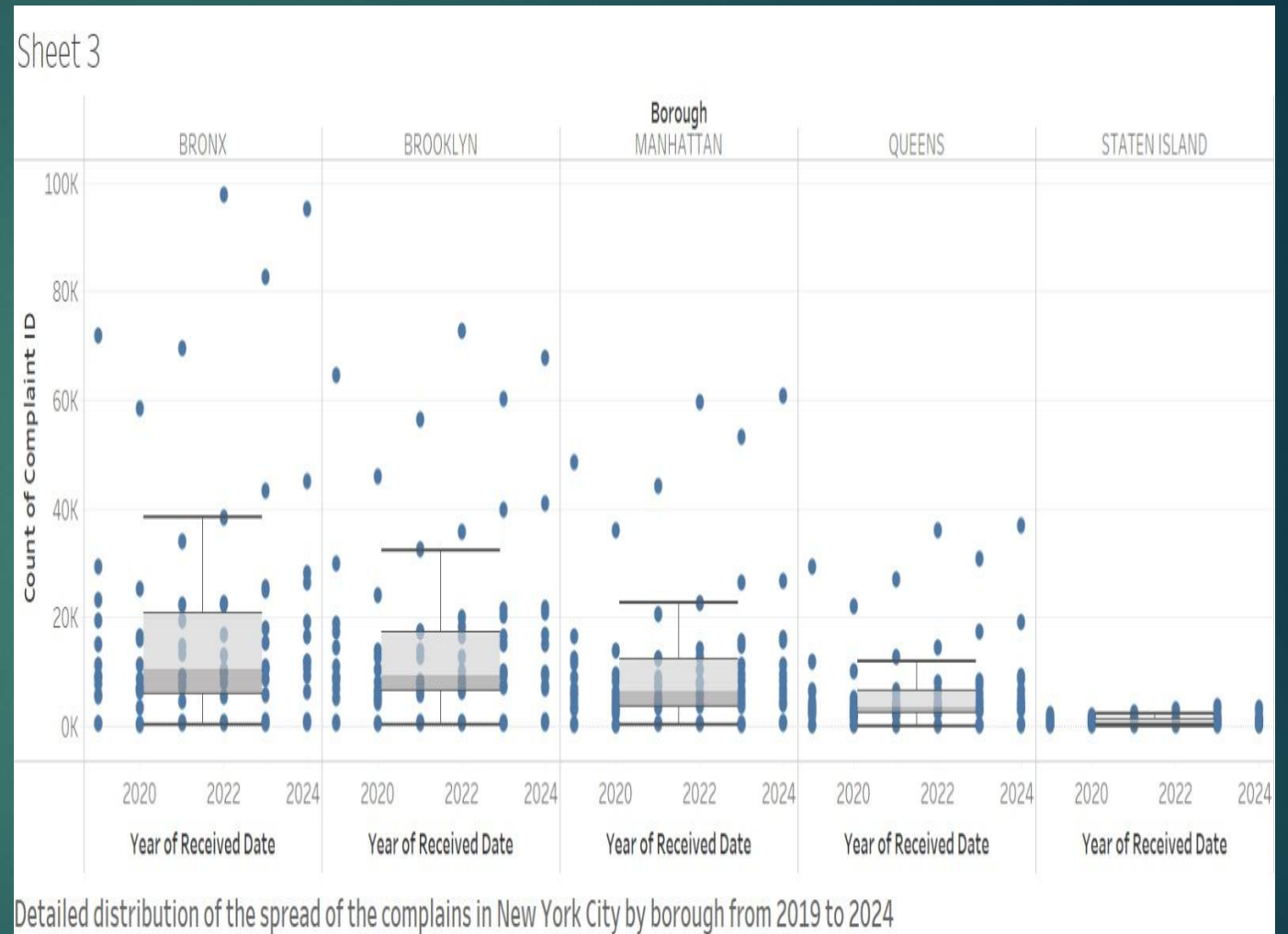


## Statistical Overview of Complaint Distribution

### *How Are Complaints Distributed Over Time?*

#### Key Insights:

- Outliers indicate occasional surges in complaints.
- Consistent median complaint levels suggest persistent issues.



# Conclusion:

## **Complaint and Violation Trends:**

Brooklyn records the highest number of complaints, but a lower percentage of them turn into violations.

The Bronx has the highest violation percentage (46%), suggesting more severe issues.

## **Resolution Efficiency:**

Queens and Staten Island resolve complaints faster, whereas the Bronx experiences prolonged resolution times, indicating systemic inefficiencies.

High-priority violations take approximately 25.5 days to resolve, but lower-priority issues can take nearly three months.

## **Geographic & Seasonal Insights:**

Complaint hotspots are concentrated in Brooklyn and the Bronx, highlighting the need for targeted enforcement.

Heat/Hot Water complaints peak in winter, suggesting a need for proactive interventions during colder months.

## **Long-Term Trends & Policy Implications:**

While the number of resolved complaints has increased significantly from 2020 to 2024, the rise in open complaints suggests a growing backlog.



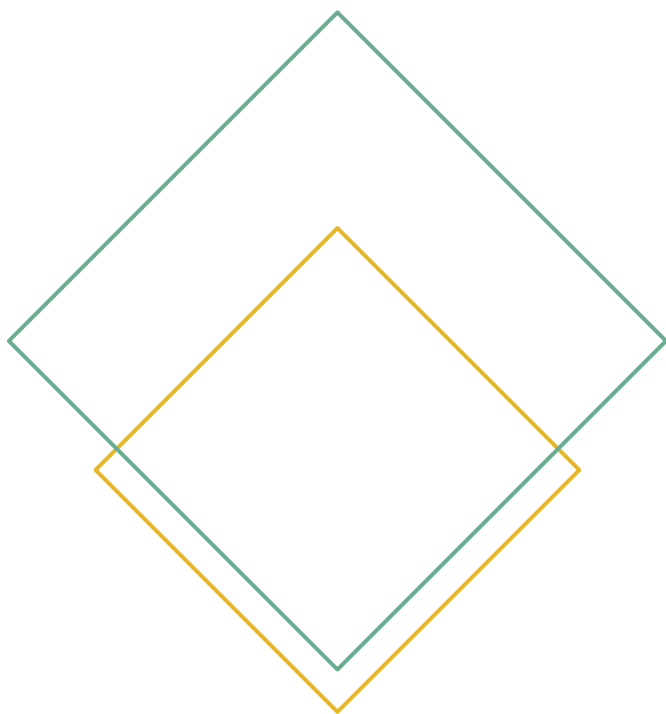
## Key Findings & Next Steps

- Brooklyn has the highest complaints but fewer violations.
- The Bronx has the worst resolution times—needs urgent action.
- Queens resolves complaints faster—best practices?
- Low-priority violations take too long—resource allocation needed.
- Complaints keep increasing yearly—what policies can fix this?



## How Can We Improve NYC Housing Conditions?

- Proposed Actions:
  - Targeted interventions in Bronx & Brooklyn.
  - Resource reallocation to reduce backlog.
  - Policy reform to streamline low-priority violations.
  - More frequent inspections to prevent issues before complaints arise.
- Final Thought:
  - *Data is a powerful tool for improving city enforcement—let's use it!*



**Thank You**